# Warrant

*Don't let the agent grade its own homework.*

别让 AI 代理给自己的作业打分。

On-chain trade limits for AI agents, a signed pre-trade disclosure, and now an independent price check: Chainlink Data Feeds re-price every disclosure on chain, and an agent that signs PASS on a bad quote is caught and loses reputation.

链上交易授权 + 交易前签名披露 + 新增独立价格复核：Chainlink 在链上重算每一份披露，对坏报价签 PASS 的代理会被当场揭穿并扣信誉。

# An AI agent's trading limits, on chain and checkable.

把 AI 代理的交易边界放上链，让任何人都能核查。

01
### Grant
授权人设定单笔上限、滑点、价格冲击、费用等，可随时撤销。

`WarrantRegistry`

02
### Preflight
代理下单前，按链上授权给真实报价打分。

`mm warrant preflight`

03
### Disclose
代理钱包签名打分结论，任何人可代为上链。

`EIP-712 → WarrantAttestor`

04
### Reputation
每份已验证的披露写成 ERC-8004 信誉记录。

`WarrantReputation`

05 · NEW
### Price check
Chainlink 独立重算报价，核对代理说的是不是真话。

`WarrantPriceCheck`

**Arc**
mainnet, chain 5042
已上主网

**ERC-8004**
Arc's own registries, agent 1419
接入 Arc 官方注册表

**78 + 65**
contract + plugin tests, all passing
合约与插件测试全部通过

**MIT**
open source, every tx linked in README
开源，每笔交易可追溯

*Steps 01–04 were built before this event (Monad Metropolis, Avalanche Buildathon, Arc Microgrants); step 05 is the increment for ETH Shenzhen 2026.* 01 – 04 为既有工作，05 为本届新增。

# Every number on the scorecard came from the agent.

打分卡上的每个数字，原本都由代理自己算。

```
// what the agent signs (EIP-712)
PreflightAttestation {
  principal      0x3fd7…0f73
  agent          0x299e…0e0e
  quoteHash      keccak(quote id)
  warrantVersion 1
  verdict        PASS
  scorecardHash  0x…
}

// checked on chain: signer = agent,
// warrant active at that version.
// NOT checked: was PASS true?
```

**A modified plugin can print PASS on anything.**

插件被改过，任何报价都能显示 PASS。

**An agent can skip preflight and sign PASS anyway.**

代理可以不打分，直接签 PASS。

**A bad quote can come with flattering price data.**

坏报价也可能附带好看的价格数据。

**Until now, the chain could not tell.**

此前，链上无从分辨。

*Source: `contracts/src/WarrantAttestor.sol` (`submit` verifies the signer and the warrant version only).*

# Chainlink re-prices it. The contract checks the verdict.

Chainlink 重算报价，合约核对代理的结论。




| AGENT DISCLOSED | WITHIN BUDGET | OVER BUDGET, INSIDE ORACLE TOLERANCE | OUT OF BOUNDS |
|---|---|---|---|
| PASS 报通过 | **consistent +1** | **consistent +1** (plugin warns) | **CONTRADICTED −1** 说谎 |
| WARN / FAIL 报警告/不通过 | **consistent +1** | **consistent +1** | **consistent +1** (it told the truth) |

*Stale feed (> 25 h) or non-positive price → revert: an unpriceable quote is never recorded as checked. Tolerance 100 bps = the two feeds' 0.5% deviation thresholds. No admin; feeds fixed at deployment.* 价格源过期即回滚，不记录。

# A real quote, a real agent wallet, two Chainlink feeds.

真实报价、真实代理钱包、两个 Chainlink 价格源。

**0 bps**

shortfall: Chainlink priced 0.5 USDC at 448,542 EURC units; the quote offered 449,499.

报价比 Chainlink 公允值还高，价格复核通过；总结论仍是 FAIL（gas 占 199 bps，上限 50）。

| STEP | TRANSACTION | RESULT |
|---|---|---|
| Deploy WarrantPriceCheck<br>1,258,254 gas | `0xcc34…b956` | `status 1` |
| mm wallet signs FAIL, relayed + rated<br>real `mm swap quote` 0xc700…1a29 | `0x9c4a…02be` | `preflight / fail` |
| checkAndRate<br>fair 448,542 · quoted 449,499 | `0xfcce…4117` | `consistent` |
| Staged dishonest agent signs PASS<br>quote 5% below Chainlink-fair | `0xbde8…8f4d` | `contradicted` |

*Source: receipts and `PriceChecked` events read from rpc.mainnet.arc.io, 2026-10-14. Feeds: USDC/USD `0x84EA…7905`, EURC/USD `0x361b…Bf23`. The dishonest agent is staged and labelled so in its ERC-8004 registration. No swap was executed.*

# Paid only if it delivered what it disclosed.

ERC-8183 托管任务：交付量对得上自己的披露和 Chainlink 公允价，才拿得到报酬。

# PAID

Job 1 on Arc mainnet: agent 1419 swapped 0.5 USDC → 448,903 EURC units (quote 448,903; Chainlink-fair 448,542). Warrant's first real trade.

结算全看链上数据：对 Chainlink 少 0 bps（上限 450），对披露报价滑点 0 bps（上限 100）。任何人都能触发结算。

| RULE IN WARRANTSWAPJOB.SETTLE | IF BROKEN |
|---|---|
| Submit must carry the agent's on-chain Warrant disclosure for this order<br>signed by the provider; principal = client; same tokens and amount | cannot submit |
| Delivered ≥ Chainlink-fair − (impact + fee budget + tolerance) | rejected |
| Delivered ≥ disclosed quote − warrant slippage<br>ties the disclosure (quote) to the execution | rejected |
| Disclosed FAIL, traded anyway · warrant revoked or changed | rejected |
| All pass | fee → agent |

*Either way the delivered tokens go to the client. Settle `0xc512…43a3`; swap `0xeca1…95fa`; core = ERC-8183 reference implementation (MIT, unmodified). Also live: `WarrantValidator` answers ERC-8004 validation requests (agent 1419: 100).* 本页数据均为 Arc 主网链上读数，2026-10-14。

# Same bad quote. One agent tells the truth.

同一笔坏报价，一个代理说真话，一个说谎。

## Honest agent 诚实代理

```
$ mm warrant preflight --price-check …
fair     448542 EURC units
quoted   426114
X Chainlink price check
  500 bps below fair · limit 200 (+100)

signs FAIL → relayed → checkAndRate
PriceChecked
  out of bounds   true
  contradicted    false

ERC-8004  consistent 1 · contradicted 0
```

## Dishonest agent 说谎代理

```
$ # skips preflight
fair     448542 EURC units
quoted   426114
  (never looked)


signs PASS → relayed → checkAndRate
PriceChecked
  out of bounds   true
  contradicted    true

ERC-8004  consistent 0 · contradicted 1
```

*Output of `node scripts/demo-pricecheck.mjs`: an Arc mainnet fork with the deployed contracts and live feeds; both agents and the quote are staged.* 本地分叉 Arc 主网运行，零成本，可现场复现。

# Reveal, not enforce. Now with a second opinion.

只揭示，不拦截；现在多了一个独立的第二意见。

**Only tokens with a feed are checkable (on Arc: USDC, EURC).**
只能复核有 Chainlink 价格源的币种。

**A 100 bps tolerance absorbs oracle lag; smaller lies pass.**
100 bps 容差吸收价格源滞后，更小的偏差不会被标记。

**The check runs when someone calls it; unchecked is visible, not penalised.**
有人调用才复核；未复核可见，但不扣分。

**Gas outside the quote is not in the output; the fee-load check covers it.**
报价外的 gas 不计入，由费用占比一项覆盖。

**Mainnet, not audited; only the job contracts escrow funds, between submit and settle.**
已上主网，未审计；只有托管任务在提交到结算之间托管资金。

CODE · 代码

## github.com/Beltran12138/warrant

RUN THE DEMO · 复现演示

```
npm install && npm run build
node scripts/demo-pricecheck.mjs
```

BUILT FOR THIS EVENT · 本届新增

WarrantPriceCheck, WarrantValidator (ERC-8004), WarrantSwapJob (ERC-8183); 45 contract and 6 plugin tests; all live on Arc mainnet.

独立开发，使用 AI 编程工具，已在 README 披露。